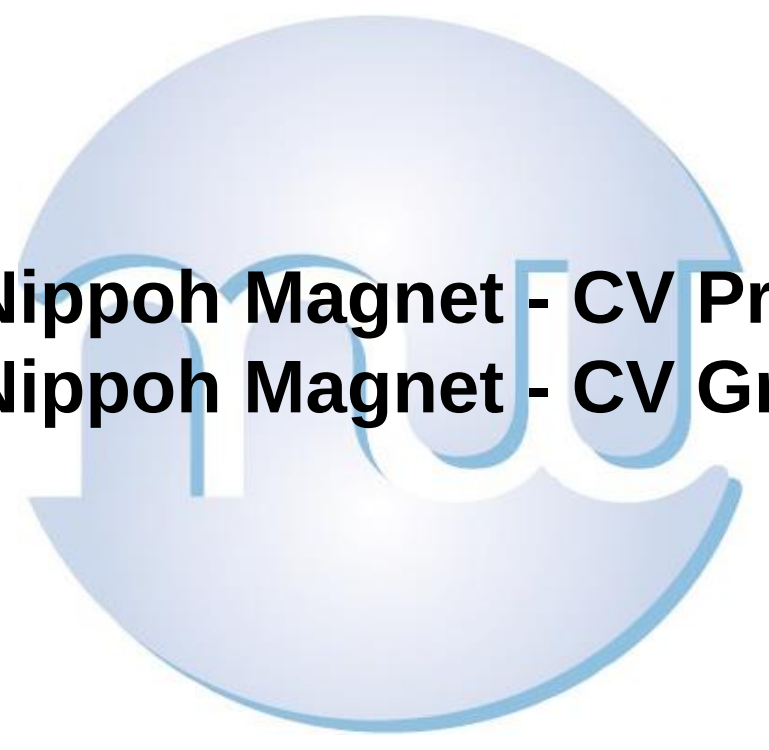


Business Trip Report

Date: 12 Mar

- 
- A. Nippoh Magnet - CV Press**
 - B. Nippoh Magnet - CV Grinding**

**Team : Mr Tan (TS Fabricator)
Mr Siva (Maruwa)**

CONFIDENTIAL

Discussion details

Date: 12 Mar

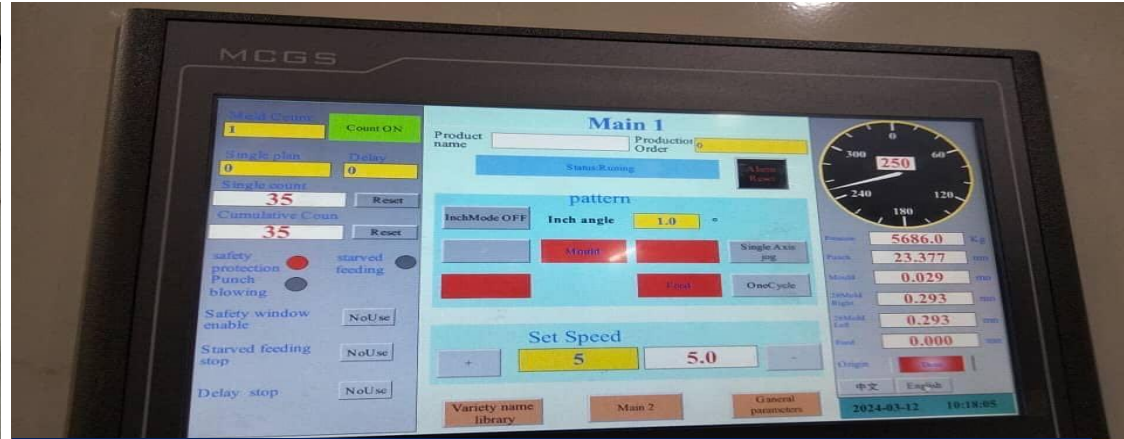
Time : 9:00am

Nippoh Magnet - CV 20ton Press

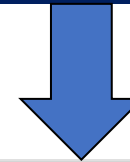
**Team : Mr Tan (TS Fabricator)
Mr Siva (Maruwa)**

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Safety	Safety control for machine	Available	lot Connectable
Machina	Servo system	Mr Tan	Maker : Italy (OEM in China)
	Body	Iron	
	Installation	Complete in China before shipouty	
	Cyclinder	Auto greasing - timer	
	Mechanisma	Good - Silent type pressing	
	Pressure capacity	un by torque - Ball screw	Japan NSK
	Pressure adjustment	Can be done	
	Dust control - machine cleanness	Not applicable	Propose for vacuum
	Cooling system	Not Applicable	
	Lifespan	5 years	Base from Supplier feedback
	Stroke	Can control	Timing
Tooling	Current tool fixable	Flexible fixing	
Automation	Auto Pick up	Uniform with other machines Auto weight check - freq decide by Maruwa	3 type sample video provided
		System for cleaning & setter arrangement available	
IoT	Output signal - Key info	no problem	
Sample	Press sample	Ongoing - 60pcs (Firing)	Handcarry by Siva
	Consistency	Stable but need to confirm more sam	R&R test
Training	Machine usage method		
	Routine maintenance	Mr Tan	
Sales	Technical support	Malaysia	
Delivery	Leadtime	2 months- 30th May	Exclude automation
Capacity		1 min : 20pcs	Targetted - 22kpcs/day/machine
Power Supply		3phase 360 ~415	Need to discuss
Powder		20kg powder to ship	Final confirmation



Supplier capable of doing their own automations
Pick up automation saved this location



This PC > maruwaroot (\\mmsbsql27) (F:) > winapps > CHINA MACHINE > CV PRESS MACHINE

Name	Date modified	Type	Size
Press Machine Video #3	3/12/2024 11:07 PM	MP4 File	6,168 KB
Sample Press Automation #3	3/12/2024 10:21 PM	MP4 File	13,999 KB
Sample Press Automation #1	3/12/2024 10:20 PM	MP4 File	8,388 KB
Sample Press Automation #2	3/12/2024 10:20 PM	MP4 File	19,024 KB
Press Machine Video #2	3/12/2024 11:07 PM	MP4 File	3,430 KB
Press Machine Video #1	3/12/2024 11:07 PM	MP4 File	4,308 KB

Discussion details

Date: 12 Mar

Time : 2:30pm

Nippoh Magnet - CV Grinding Machine

**Team : Mr Tan (TS Fabricator)
Mr Siva (Maruwa)**

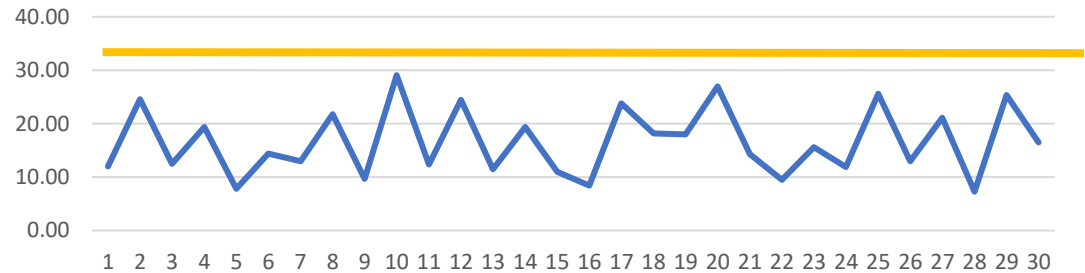
CONFIDENTIAL

Topic	Detail	Remarks	
Safety	Safety control for machine	Required - compulsory	
Machine	Hydraulic system	Air spindle - not relevant to hydraulic	
	Installation	Required video - installation	
	Spindle	Greasing manually	CMMS
	Maximum grinding diameter	700D	
	Precision level	10micron	
	Pressure capacity	Adjustable	Different
	Thickness adjustment	probe	
	Sludge control system	Magnetic separator with Cooling system	
	Coolant system	Cooling water	
	Lifespan		
	Capacity	Will re-confirm with new set of samples	Earlier was pretesting
	Product loading method	Video	
Grinding tools	Wheels grade	#320 => #270	Ra & Flatness Confirm top & bottom wheel - study with various study
	wheel change time	Depend	
	wheel life span	Depend	
	Carrier	Standard 10pcs /carrier or will effect flatness	
Automation	Auto Pick up / loading	Uniform with other machines	
IoT	Output signal - Key info	PLC communication	Model will be provided to Maruwa
Sample	Grinding sample	Additional sample of 120pcs	Powder will be sent out by 18/03
	Consistency Flatness	Re-check again	
Training	Machine usage method	Training slide- Mr Tan	
	Routine maintenance	Training Slide - Mr Tan	
Sales	Technical support		
Delivery	Leadtime	ETD :30th Apr	

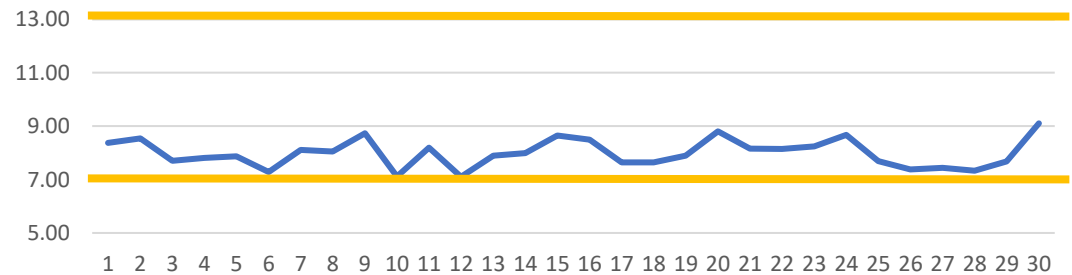
Samples Flatness & Roughness Grinding Machine China

No.	Flatness	Roughness	Thickness	Small t
	(u")	Ra	(mm)	(mm)
Min	0	7	3.835	1.829
Centre	16.5	10	3.938	1.931
Max	33	13	4.04	2.032
1	12.00	8.38	3.822	1.785
2	24.60	8.54	3.824	1.784
3	12.50	7.70	3.822	1.786
4	19.40	7.81	3.820	1.789
5	7.80	7.87	3.819	1.813
6	14.40	7.29	3.820	1.807
7	13.00	8.11	3.820	1.848
8	21.80	8.05	3.821	1.847
9	9.70	8.74	3.823	1.818
10	29.10	7.12	3.822	1.808
11	12.40	8.19	3.818	1.826
12	24.50	7.10	3.819	1.784
13	11.50	7.90	3.825	1.854
14	19.40	7.99	3.819	1.862
15	11.00	8.65	3.824	1.827
16	8.40	8.50	3.820	1.844
17	23.80	7.65	3.821	1.827
18	18.20	7.65	3.821	1.780
19	18.00	7.90	3.823	1.835
20	27.00	8.81	3.820	1.835
21	14.30	8.16		
22	9.50	8.15		
23	15.60	8.24		
24	11.90	8.68		
25	25.60	7.69		
26	13.00	7.38		
27	21.10	7.44		
28	7.30	7.33		
29	25.40	7.68		
30	16.50	9.11		
Avg	16.62	7.99	3.82	1.82
Max	29.100	9.110	3.825	1.862
Min	7.300	7.100	3.818	1.780

FLATNESS TREND - SAMPLE



ROUGHNESS TREND - SAMPLE



Rediscuss with supplier to redo sample on the wheel diamond grade

Current Ra → Minimum Action

Re-test with diamond grade from #320 → #300 → #270

***Target : Eliminate the Lapping & washing process**

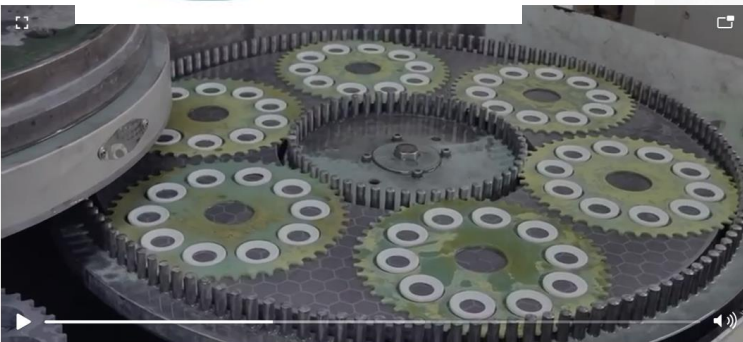


New Grinding Machine
Capacity : 17k/day
: 510kpcs/mth
Supplier will redo the capacity
check with additional sample



Sample Grinding stone

Supplier propose loading & unloading
method (Human dependent)
Sample video



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Name	Date modified	Type	Size
CV Grinding Loading Method	3/12/2024 10:18 PM	MP4 File	33,232 KB
CV Grinding machine	3/6/2024 5:07 PM	MP4 File	15,671 KB



Thank You